24MCAL191 – PROGRAMMING LAB

Report Submitted By

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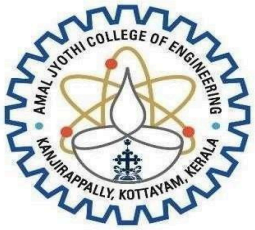
In Partial fulfilment for the Award of the Degree Of

MASTER OF COMPUTER APPLICATIONS
(MCA TWO YEARS)
[Accredited by NBA]
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY



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KANJIRAPPALLY

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2025-2026

DEPARTMENT OF COMPUTER APPLICATIONS

AMAL JYOTHI COLLEGE OF ENGINEERING

KANJIRAPALLY

CERTIFICATE

This is to certify that the lab report, "**24MCAL191 - PROGRAMMIMG LAB"** is the bona fide work of **JOSEPH MATHEW AJC25MCA-2041** in partial fulfilment of the requirements for the award of the Degree of Master of Computer Applications under APJ Abdul Kalam Technological University, during the year **2025-26.**

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Course Code	Course Name	Syllabus Year
24MCAL191	PROGRAMMIMG LAB	2024

VISION

To promote an academic and research environment conducive for innovation centric technical education.

MISSION

● MS1 - Provide foundations and advanced technical education in both theoretical and applied Computer Applications in-line with Industry demands.
● MS2 - Create highly skilled computer professionals capable of designing and innovating real life solutions.
● MS3 - Sustain an academic environment conducive to research and teaching focused to generate up-skilled professionals with ethical values.
● MS4 - Promote entrepreneurial initiatives and innovations capable of bridging and contributing with sustainable, socially relevant technology solutions.

COURSE OUTCOME

CO	Outcome
CO1	Understands basics of Python Programming language including input/output functions, operators, basic and collection data types.
CO2	Implement decision making, looping constructs and functions.
CO3	Design modules and packages - built in and user defined packages.
CO4	Implement object-oriented programming and exception handling.
CO5	Create files and form regular expressions for effective search operations on strings and files.

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Experiment No: 1

Aim: Generate a list containing only the positive numbers from a given list of integers.

CO1: Understands basics of Python Programming language including input/output functions, operators, basic and collection data types.

Program code:

```
numbers = list(map(int, input("Enter integers separated by space: ").split()))
positive = []
for n in numbers:
if n > 0:
positive.append(n)
print(positive)
```

Result: The program was executed. The result was obtained successfully. Thus CO1 was obtained.

Output:

```
Enter integers separated by space: 2 4 6 -1 -3 0
[2, 4, 6]
```

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Experiment No: 2

Aim: Compute the square of N numbers.

CO1: Understands basics of Python Programming language including input/output functions, operators, basic and collection data types.

Program code:

```
n = int(input("Enter number of elements: "))
numbers = []

print("Enter the numbers:")
for i in range(n):
    num = int(input())
    numbers.append(num)

squares = []
for num in numbers:
    squares.append(num * num)

print("Squares of the numbers:", squares)
```

Result: The program was executed. The result was obtained successfully. Thus CO1 was obtained.

Output:

```
Enter number of elements: 4
Enter the numbers:
2
3
4
5
Squares of the numbers: [4, 9, 16, 25]
```

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Experiment No: 3

Aim: Form a list containing all vowels selected from a given word.

CO2: Implement decision making, looping constructs and functions.

Program code:

```
word = input("Enter a word: ")

vowels = ['a', 'e', 'i', 'o', 'u']
vowel_list = []

for ch in word.lower():
    if ch in vowels:
        vowel_list.append(ch)

print("Vowels in the word:", vowel_list)
```

Result: The program was executed. The result was obtained successfully. Thus CO2 was obtained.

Output:

```
Enter a word: Education
Vowels in the word: ['e', 'u', 'a', 'i', 'o']
```

Experiment No: 4

Aim: Store a list of first names and count the total occurrences of the letter ‘a’ in the list.

CO2: Implement decision making, looping constructs and functions.

Program code:

```
n = int(input("Enter number of names: "))
names = []

print("Enter the first names:")
for i in range(n):
    name = input()
    names.append(name)

count = 0
for name in names:
    count += name.lower().count('a')

print("Total occurrences of 'a':", count)
```

Result: The program was executed. The result was obtained successfully. Thus CO2 was obtained.

Output:

```
Enter number of names: 3
Enter the first names:
thomas
jacob
mathew
Total occurrences of 'a': 3
```

Experiment No: 5

Aim: Enter two lists of integers and check:

a) whether the lists are of equal length,

b) whether their sums are equal, and

c) whether any values are common to both lists.

CO2: Implement decision making, looping constructs and functions.

Program code:

```
list1 = list(map(int, input("Enter first list: ").split()))
list2 = list(map(int, input("Enter second list: ").split()))

# a) Equal length
if len(list1) == len(list2):
print("Same length")
else:
print("Different length")

# b) Equal sum
if sum(list1) == sum(list2):
print("Same sum")
else:
print("Different sum")

# c) Common values
common = []
for x in list1:
if x in list2 and x not in common:
common.append(x)

if common:
print("Common values:", common)
```

```
else:
print("No common values")
```

Result: The program was executed. The result was obtained successfully. Thus CO2 was obtained.

Output:

```
Enter first list: 2 4 5 6 7
Enter second list: 6 4 2 1 3
Same length
Different sum
Common values: [2, 4, 6]
```

Experiment No: 6

Aim: Create a new string in which all occurrences of the first character are replaced with '$', except the first character itself.

CO2: Implement decision making, looping constructs and functions.

Program code:

```
s = input("Enter a string: ")

first = s[0]
new_string = first

for ch in s[1:]:
if ch == first:
new_string += '$'
else:
new_string += ch

print("New string:", new_string)
```

Result: The program was executed. The result was obtained successfully. Thus CO2 was obtained.

Output:

```
Enter a string: restart
New string: resta$t
```

Experiment No: 7

Aim: Create a string from a given string by exchanging its first and last characters.

CO2: Implement decision making, looping constructs and functions.

Program code:

```
string = input("Enter a string: ")

if len(string) < 2:
   print("New string:", string)
else:
   new_string = string[-1] + string[1:-1] + string[0]
   print("New string:", new_string)
```

Result: The program was executed. The result was obtained successfully. Thus CO2 was obtained.

Output:

```
Enter a string: joseph
New string: hosepj
```

Experiment No: 8

Aim: Print all colors that are present in color_list1 but not in color_list2.

CO2: Implement decision making, looping constructs and functions.

Program code:

```
color_list1 = ["Red", "Green", "White", "Black"]
color_list2 = ["Green", "White"]

for color in color_list1:
    if color not in color_list2:
        print(color)
```

Result: The program was executed. The result was obtained successfully. Thus CO2 was obtained.

Output:

```
Enter colors of list 1: red blue yellow green
Enter colors of list 2: red blue
Colors in list1 but not in list2:
yellow
green
```

Experiment No: 9

Aim: Create a single string separated by a space from two strings after swapping the character at index 1.

CO2: Implement decision making, looping constructs and functions.

Program code:

```
s1 = input("Enter first string: ")
s2 = input("Enter second string: ")
if len(s1) > 1 and len(s2) > 1:
s1_new = s1[0] + s2[1] + s1[2:]
s2_new = s2[0] + s1[1] + s2[2:]
else:
s1_new = s1
s2_new = s2
result = s1_new + " " + s2_new
print("Result:", result)
```

Result: The program was executed. The result was obtained successfully. Thus CO2 was obtained.

Output:

```
Enter first string: joseph
Enter second string: thomas
Result: jhseph toomas
```

Experiment No: 10

Aim: Sort a dictionary in both ascending and descending order.

CO2: Implement decision making, looping constructs and functions.

Program code:

```
my_dict = {
   'b': 7,
   'a': 1,
   'd': 3,
   'c': 8
}

asc_dict = dict(sorted(my_dict.items()))
desc_dict = dict(sorted(my_dict.items(), reverse=True))

print("Ascending order:", asc_dict)
print("Descending order:", desc_dict)
```

Result: The program was executed. The result was obtained successfully. Thus CO2 was obtained.

Output:

```
Ascending order: {'a': 1, 'b': 7, 'c': 8, 'd': 3}
Descending order: {'d': 3, 'c': 8, 'b': 7, 'a': 1}
```

Experiment No: 11

Aim: Merge two dictionaries into one.

CO2: Implement decision making, looping constructs and functions.

Program code:

```
d1 = eval(input("Enter first dictionary: "))
d2 = eval(input("Enter second dictionary: "))

d1.update(d2)
print("Merged dictionary:", d1)
```

Result: The program was executed. The result was obtained successfully. Thus CO2 was obtained.

Output:

```
Enter first dictionary: {'a':3, 'f':4}
Enter second dictionary: {'e':1, 'h':5}
Merged dictionary: {'a': 3, 'f': 4, 'e': 1, 'h': 5}
```

Experiment No: 12

Aim: Find the greatest common divisor (GCD) of two numbers.

CO2: Implement decision making, looping constructs and functions.

Program code:

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

while b != 0:
    a, b = b, a % b
print("GCD:", a)
```

Result: The program was executed. The result was obtained successfully. Thus CO2 was obtained.

Output:

```
Enter first number: 12
Enter second number: 24
GCD: 12
```

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Experiment No: 13

Aim: From a list of integers, create a new list after removing all even numbers.

CO2: Implement decision making, looping constructs and functions.

Program code:

```
numbers = list(map(int, input("Enter integers separated by space: ").split()))

odd_numbers = []

for num in numbers:
if num % 2 != 0:
odd_numbers.append(num)

print("List after removing even numbers:", odd_numbers)
```

Result: The program was executed. The result was obtained successfully. Thus CO2 was obtained.

Output:

```
Enter integers separated by space: 2 3 7 5 6 1 9
List after removing even numbers: [3, 7, 5, 1, 9]
```

Experiment No: 14

Aim: Generate a Fibonacci series of N terms.

CO2: Implement decision making, looping constructs and functions.

Program code:

```
N = 11
a, b = 0, 1
fib_series = []

for _ in range(N):
    fib_series.append(a)
    a, b = b, a + b

print(fib_series)
```

Result: The program was executed. The result was obtained successfully. Thus CO2 was obtained.

Output:

```
[0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55]
```

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Experiment No: 15

Aim: Find the sum of all items in a list using function.

CO3: Design modules and packages - built in and user defined packages

Program code:

```
def sum_of_list(lst):
    total = 0
    for num in lst:
        total += num
    return total

numbers = list(map(int, input("Enter the numbers: ").split()))
print("Sum of all items:", sum_of_list(numbers))
```

Result: The program was executed. The result was obtained successfully. Thus CO3 was obtained.

Output:

```
Enter the numbers: 3 4 5 6 7
Sum of all items: 25
```

Experiment No: 16

Aim: Count the frequency of each character in a string using function.

CO3: Design modules and packages - built in and user defined packages

Program code:

```
def char_frequency(s):
    freq = {}
    for ch in s:
        if ch in freq:
            freq[ch] += 1
        else:
            freq[ch] = 1
    return freq

text = str(input("enter the string:"))
result = char_frequency(text)
print(result)
```

Result: The program was executed. The result was obtained successfully. Thus CO3 was obtained.

Output:

```
enter the string:python
{'p': 1, 'y': 1, 't': 1, 'h': 1, 'o': 1, 'n': 1}
```

Experiment No: 17

Aim: Add ‘ing’ to the end of a given string. If it already ends with ‘ing’, add ‘ly’ instead using function

CO3: Design modules and packages - built in and user defined packages

Program code:

```
def add_suffix(s):
    if s.endswith("ing"):
        return s + "ly"
    else:
        return s + "ing"

string = input("Enter a string: ")
result = add_suffix(string)

print("Result:", result)
```

Result: The program was executed. The result was obtained successfully. Thus CO3 was obtained.

Output:

```
Enter a string: soft
Result: softing
```

```
Enter a string: softing
Result: softingly
```

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Experiment No: 18

Aim: Write lambda functions to find the areas of a square, a rectangle, and a triangle using function

CO3: Design modules and packages - built in and user defined packages

Program code:

```
area_square = lambda side: side * side
area_rectangle = lambda length, breadth: length * breadth
area_triangle = lambda base, height: 0.5 * base * height


s = float(input("Enter side of square: "))
print("Area of square:", area_square(s))

l = float(input("Enter length of rectangle: "))
b = float(input("Enter breadth of rectangle: "))
print("Area of rectangle:", area_rectangle(l, b))

base = float(input("Enter base of triangle: "))
h = float(input("Enter height of triangle: "))
print("Area of triangle:", area_triangle(base, h))
```

Result: The program was executed. The result was obtained successfully. Thus CO3 was obtained.

Output:

```
Enter side of square: 4
Area of square: 16.0
Enter length of rectangle: 2
Enter breadth of rectangle: 4
Area of rectangle: 8.0
Enter base of triangle: 3
Enter height of triangle: 5
Area of triangle: 7.5
```

Experiment No: 19

Aim: Create a Rectangle class with attributes length and breadth,along with methods to compute area and perimeter.Compare two rectangle objects based on their areas.

CO4: Implement object-oriented programming and exception handling.

Program code:

```
class Rectangle:
    def __init__(self, length, breadth):
        self.length = length
        self.breadth = breadth

    def area(self):
        return self.length * self.breadth
    def perimeter(self):
        return 2 * (self.length + self.breadth)


r1 = Rectangle(10,10)
print(f"Area of Rectangle: {r1.area()}")
print(f"Perimeter ofRectangle Perimeter:{r1.perimeter()}")
r2 = Rectangle(10, 20)
print(f"Area of Rectangle: {r2.area()}")
print(f"Perimeter ofRectangle Perimeter:{r2.perimeter()}")

if r1.area() > r2.area():

    print("\nRectangle 1 has larger area")
elif r1.area() < r2.area():
    print("\nRectangle 2 has larger area")
else:
    print("\nBoth rectangles have equal area")
```

Result: The program was executed. The result was obtained successfully. Thus CO4 was obtained.

Output:

```
Area of Rectangle: 100
Perimeter ofRectangle Perimeter:40
Area of Rectangle: 200
Perimeter ofRectangle Perimeter:60

Rectangle 2 has larger area
```

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Experiment No: 20

Aim: Create a BankAccount class with attributes: accountnumber, name, account type, and balance. Write constructor and methods to deposit and withdraw money from the bank account

CO4: Implement object-oriented programming and exception handling.

Program code:

```
class BankAccount:
    def __init__(self):
        self.balance = 10000


    def deposit(self, amount):
        self.balance += amount
        return self.balance
    def withdraw(self, amount):
        self.balance -= amount
        return self.balance

a = BankAccount()
print(a.deposit(500))
print(a.withdraw(1500))
print(a.deposit(1000))
```

Result: The program was executed. The result was obtained successfully. Thus CO4 was obtained.

Output:

```
10500
9000
10000
```

.

Experiment No: 21

Aim: Create a Time class with private attributes hour, minute, and second, and overload the + operator to add two time objects.

CO4: Implement object-oriented programming and exception handling.

Program code:

```
class Time:
    def __init__(self, h, m, s):
        self.__hour = h
        self.__minute = m
        self.__second = s

    def __add__(self, other):
        s = self.__second + other.__second
        m = self.__minute + other.__minute
        h = self.__hour + other.__hour

        if s >= 60:
            m += s // 60
            s = s % 60

        if m >= 60:
            h += m // 60
            m = m % 60

        return Time(h, m, s)

    def display(self):
        print(f"{self.__hour:02d}:{self.__minute:02d}:{self.__second:02d}")

print("Enter Time 1")
h1 = int(input("Hour: "))
m1 = int(input("Minute: "))
s1 = int(input("Second: "))
```

```
print("Enter Time 2")
h2 = int(input("Hour: "))
m2 = int(input("Minute: "))
s2 = int(input("Second: "))

t1 = Time(h1, m1, s1)
t2 = Time(h2, m2, s2)

t3 = t1 + t2

print("Sum of Time:")
t3.display()
```

Result: The program was executed. The result was obtained successfully. Thus CO4 was obtained.

Output:

```
Enter Time 1
Hour: 2
Minute: 30
Second: 24
Enter Time 2
Hour: 3
Minute: 23
Second: 10
Sum of Time:
05:53:34
```

Experiment No: 22

Aim: Write a Python program to read a file line by line and store each line in a list.

CO5: Create files and form regular expressions for effective search operations on strings and files.

Program code:

```
lines = []

with open("sample.txt", "r") as file:
    for line in file:
        lines.append(line.strip())

print("Lines stored in list:")
print(lines)
```

Result: The program was executed. The result was obtained successfully. Thus CO5 was obtained.

Output:

```
Lines stored in list:
['Apple', 'Banana', 'Mango']
```

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Experiment No: 23

Aim: Write a Python program to read specific columns of a CSV file and print the contents.

CO5: Create files and form regular expressions for effective search operations on strings and files.

Program code:

Data.csv

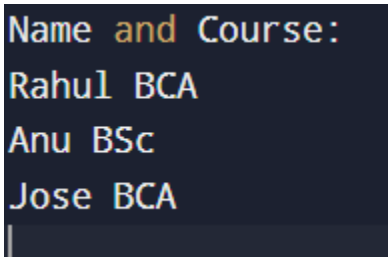
```
ID,Name,Age,Course
1,Rahul,20,BCA
2,Anu,21,BSc
3,Jose,22,BCA
```

```
import csv

with open("data.csv", "r", newline="") as file:
    reader = csv.reader(file)
    header = next(reader)
    print("Name and Course:")
    for row in reader:
        if len(row) >= 4:
            print(row[1], row[3])
```

Result: The program was executed. The result was obtained successfully. Thus CO5 was obtained.

Output:



Experiment No: 24

Aim: Write a Python program to write a dictionary to a CSV file,then read back the file and display its contents.
CO5: Create files and form regular expressions for effective search operations on strings and files.

Program code:

```
import csv

data = {
   "Name": "Alice",
   "Age": 25,
   "City": "New York",
   "Profession": "Engineer"
}

filename = "data.csv"

with open(filename, mode="w", newline="") as file:
   writer = csv.writer(file)
   writer.writerow(["Key", "Value"])
   for key, value in data.items():
      writer.writerow([key, value])

print("Dictionary written to CSV file.\n")

print("Contents of CSV file:")
with open(filename, mode="r") as file:
   reader = csv.reader(file)
   for row in reader:
      print(row)
```

Result: The program was executed. The result was obtained successfully. Thus CO5 was obtained.

Output:

```
Contents of CSV file:
['Key', 'Value']
['Name', 'Alice']
['Age', '25']
['City', 'New York']
['Profession', 'Engineer']
```